CHAPTER

02

Sensation and Perception

All psychological life begins with knowing what is happening around us. **Sensation** and **perception** supply the raw data for learning, memory, thinking, emotion and self-awareness. Without them, a person would be like someone in a coma. The module opens with a story: a couple travelling with a horse is criticized whether they ride together, ride one at a time, or walk beside the horse. The **same situation** is perceived differently by different people. This chapter explains how we detect the world (sensation), how we make sense of it (perception), and why our perceptions can differ, and even deceive us.

LESSONS IN THIS CHAPTER

2.1 The Meaning of Sensation and Perception

2.2 The Sensory Laws: Thresholds and Adaptation

2.3 Selectivity of Perception: Attention

2.4 Form Perception and Gestalt Organization

2.5 Depth Perception, Constancy and Illusion

Freshman A+ · General Psychology (Psyc 1011)

CHAPTER 2 · LESSON 2.1

The Meaning of Sensation and Perception

AFTER THIS LESSON YOU CAN

- Define sensation, stimulus and transduction
- Define perception and explain how it differs from sensation
- Explain why sensation and perception cannot be fully separated
- Analyse the prosopagnosia case

2.1.1 Sensation

D SENSATION

Sensation is the process in which **stimulation of receptor cells** (in the eyes, ears, nose, tongue and skin) sends **nerve impulses to the brain**. It **detects** a stimulus from the body or the environment. Colour, brightness, the pitch of a tone and a bitter taste are sensations.

- **Stimulus:** a form of **energy** (light waves, sound waves, chemicals, pressure) that can affect a sense organ.
- **Transduction:** the conversion of physical energy into **neural impulses** the brain can use (e.g. the retina turns light into nerve signals).

2.1.2 Perception

D PERCEPTION

Perception is the process that **organizes and interprets** sensations, giving them **order and meaning**. It is "the act of knowing through sensation".

	Sensation	Perception
Key word	**detection**, **stimulation**	**interpretation**, **organization**
Happens in	the sense organs and nerves	the brain
Example	seeing colours, hearing sounds, detecting the black marks on this page	recognizing a melody, a face, or the letters and words on this page
Saying	"you **hear**..." / "you **watch**..."	"...but you don't **listen**" / "...but you don't **see**"

★ "LARGELY", NOT "COMPLETELY"

Hearing sounds is **largely** sensory, and forming a melody is **largely** perceptual. In real life the two cannot be separated: as soon as the brain receives sensations it automatically interprets them, and **perception cannot occur without sensation**. Sensation without meaning is just raw data: information captured by a sense organ **has no meaning by itself**.

S OLIVER SACKS: THE MAN WHO MISTOOK HIS WIFE FOR A HAT

A patient of the neurologist **Oliver Sacks** had brain damage that caused **prosopagnosia**, the inability to recognize faces. He could recognize people by their voices, and his eyesight was fine (he could see a pin on the floor). Yet he patted fire hydrants thinking they were children's heads, and grabbed his wife's head mistaking it for a hat. **Lesson:** his **sensation** was intact (he detected the facial features), but his **perception** failed (he could not organize them into a recognizable face).

SENSATION AND PERCEPTION IN DAILY LIFE

At a busy bus station in Addis Ababa, your ears receive engines, horns, shouting conductors and music (**sensation**). Among them you recognize a conductor calling "Piassa! Piassa!" and you understand where that minibus is going (**perception**). A visitor who does not speak Amharic receives exactly the same sounds but perceives only noise.

Exam Practice

10 QUESTIONS · LESSON 2.1

1 Addis Ababa University · Mid 2014

While sensation involves **__, perception involves ____**.

A) organization; coordination
B) identification; translation
C) stimulation; interpretation
D) comprehension; information

2 Ambo University · Mid 2014

Sensation is to **__ as perception is to ____**.

A) detection; transduction
B) interpretation; transmission
C) attention; organization
D) detection; interpretation

3 Arsi University · Mid 2015

Perception is the process by which we receive information from the environment.

True / False

4 Bule Hora University · Mid 2014

The process by which the brain organizes information and translates it into something meaningful is called sensation.

True / False

5 ASTU · Mid 2015

A physical stimulus such as light, sound or chemical energy is converted into a neural impulse through a process known as:

A) Perception
B) Transaction
C) Transduction
D) Sensation

6 ASTU · Mid 2015

Which of the following is a correct statement regarding sensation and perception?

A) Perception is the act of knowing through sensation
B) Form perception is our ability to see in a three-dimensional world
C) Perception occurs without sensation
D) Sensation gives meaning to stimuli

7 Bonga University · Mid 2014

Information captured by a sensory organ has meaning by itself.

True / False

8 Addis Ababa Science and Technology University · Mid 2015

Sensation is generally a psychological process, while perception is a biological process.

True / False

9 Mekelle University · Mid 2016

Perception is a process by which:

A) environmental stimuli are sensed
B) stimulus energy is converted into electrical signals
C) sensations are stored permanently in memory
D) sensations are assembled into meaningful patterns

10 Model question (module case)

Oliver Sacks' patient could see a pin on the floor but mistook his wife's head for a hat. This shows that:

A) perception can occur without sensation
B) sensation can be intact while perception fails
C) the patient was nearsighted
D) sensory adaptation had taken place

Answers & Explanations

1. C

stimulation; interpretation

Sensation is the stimulation (detection) of receptors; perception is the interpretation of that information.

2. D

detection; interpretation

Sensation detects stimuli; perception interprets them.

3. False

Receiving (detecting) information is sensation. Perception organizes and interprets it.

4. False

Organizing and giving meaning is perception.

5. C

Transduction

Transduction converts physical energy into neural messages.

6. A

Perception is the act of knowing through sensation

Perception depends on sensation and gives it meaning. Form perception is two-dimensional; depth perception is three-dimensional.

7. False

Raw sensations have no meaning until perception organizes and interprets them.

8. False

It is the reverse: sensation is largely physiological (receptors and nerves), and perception is the psychological meaning-making process.

9. D

sensations are assembled into meaningful patterns

Perception organizes sensations into meaningful patterns. A is sensation; B is transduction; C is memory.

10. B

sensation can be intact while perception fails

He detected the features (sensation) but could not organize them into a face (perception). The module stresses he was not nearsighted.

CHAPTER 2 · LESSON 2.2

The Sensory Laws: Thresholds and Adaptation

AFTER THIS LESSON YOU CAN

- Define absolute threshold and difference threshold (jnd)
- Apply Weber's law in calculations
- Explain the role of response bias
- Explain sensory adaptation and why it is useful

2.2.1 The absolute threshold

D ABSOLUTE THRESHOLD (LIMEN)

The **minimum amount of stimulation** a person can detect. Because it varies, it is defined as the intensity detected **50 % of the time** over many trials. If a faint sound is presented 30 times and you detect it 15 times, that intensity is your absolute threshold.

Examples: the least sugar that makes coffee taste sweet; a candle flame seen about 48 km away on a dark, clear night; an audiologist playing tones of different loudness and asking you to raise your hand whenever you hear one (testing your **absolute** threshold).

★ LOWER THRESHOLD = HIGHER SENSITIVITY

A person who detects a **weaker** stimulus has a **lower** threshold and is **more** sensitive. So "the higher the threshold, the higher the sensitivity" is **false**.

Detection depends not only on the **intensity** of the stimulus but also on the person's **physical and psychological state**. An important factor is **response bias**: how ready you are to report a stimulus. Walking at night in a neighbourhood you believe is dangerous, you are more likely to "hear" footsteps than in a neighbourhood you believe is safe.

2.2.2 The difference threshold and Weber's law

D DIFFERENCE THRESHOLD (JUST NOTICEABLE DIFFERENCE, JND)

The **minimum change** in stimulation that can be detected **50 % of the time**. Example: the extra sugar needed before coffee tastes sweeter; the extra volume needed before you notice a radio is louder.

F WEBER'S LAW

The jnd is a **constant fraction** of the original stimulus:

$$\text{jnd} = k \times \text{original intensity}$$

For weight, $k \approx 2\,\%$ (1/50): holding 50 oz you notice a change of 1 oz; holding 100 oz you need 2 oz. Weber's law works best for stimuli of **moderate** intensity.

? SOLVING A WEBER'S-LAW QUESTION (ARBA MINCH 2016)

"A person notices 1 kg added to 20 kg, but may not notice 1 kg added to 50 kg." Which law? And how much must be added to 50 kg to be noticed?

Step 1: The question is about noticing a **change**, not detecting a stimulus at all → **difference threshold** (Weber's law), not absolute threshold.

Step 2: 1 kg in 20 kg is a fraction of $1/20 = 5\,\%$. 1 kg in 50 kg is only $2\,\%$, too small for this person.

Step 3: To keep the same fraction: $\text{jnd} = 0.05 \times 50 = 2.5$ kg.

Rule: "Detect at all / minimum to notice" → absolute. "Notice a change / difference / increase" → difference threshold.

2.2.3 Sensory adaptation

D SENSORY ADAPTATION

The **decreasing responsiveness** of our senses to an **unchanging (constant) stimulus**. If a stimulus stays the same, we gradually stop noticing it. (When it applies to responding, it is also called **habituation**.)

◉ ADAPTATION EVERYWHERE

- A woman cooking **doro wat** hardly smells it, while a stranger passing her house notices the smell at once.
- A man who lives near a leather factory's waste yard stops noticing the bad smell.
- The first bite of a salty dish tastes very salty; after a few mouthfuls it does not.
- You jump into a cold pool and shiver; minutes later you say "the water is fine".
- You stop feeling your clothes on your body, or the vibration of a bumpy bus seat.

★ WHY ADAPTATION IS USEFUL

It lets us ignore **unchanging** background and stay alert to **important changes**. But we do **not** adapt completely to **extreme** stimuli such as severe pain or freezing cold, because ignoring them could be harmful or fatal. Adaptation is less sensitivity to an **unchanging** stimulus, not to a **changing** one.

! THRESHOLD AND ADAPTATION TRAPS

- Not noticing a **constant** smell, sound or pressure → **adaptation**, not threshold.
- Noticing a change in **amount** → **difference** threshold; detecting presence at all → **absolute** threshold.
- Weber's law concerns the difference threshold.
- Three conditions under which you may not sense a stimulus: it is **below the absolute threshold**, the **change is below the jnd**, or you have **adapted** to it.

Exam Practice

14 QUESTIONS · LESSON 2.2

1 Arsi University · Mid 2015

What is the phenomenon of no longer responding to a stimulus that continues at the same intensity?

A) Difference threshold
B) Sensory adaptation
C) Absolute threshold
D) Sensory stimulation

2 Addis Ababa University · Mid 2014

A woman preparing doro wat is not as sensitive to its smell as a stranger walking past her house. Her insensitivity is best explained by:

A) sensory adaptation
B) sensory input
C) sensory threshold
D) difference threshold

3 Bahir Dar University · Final 2015

A 25-year-old man living near a leather factory's waste yard stopped noticing its bad smell after repeated exposure. This is:

A) Sensory adaptation
B) Sensory overload
C) The just noticeable difference
D) Absolute threshold

4 Ambo University · Mid 2014

With the first bite of a flavourful dish, a student notices strong saltiness, but after a few mouthfuls the flavour is less pronounced. This implies:

A) Sensory threshold
B) Difference threshold
C) Sensory adaptation
D) Perceptual constancy

5 Ambo University · Mid 2014

Sensory adaptation is being less sensitive to changing stimulation.

True / False

6 ASTU · Mid 2015

Sensory adaptation allows us to notice differences in sensations and react to extreme changes in stimuli.

True / False

7 Arba Minch University · Mid 2016

In sensation, the higher the threshold, the higher the sensitivity.

True / False

8 Arba Minch University · Mid 2016

A person notices 1 kg added to 20 kg but may not sense the same 1 kg added to 50 kg. This best describes:

A) Absolute threshold
B) Difference threshold
C) Sensory adaptation
D) Sensory threshold

9 Injibara University · Mid 2014

You must increase the volume of your tape recorder by a certain amount before you notice a change. This refers to the law of:

A) Absolute threshold
B) Sensory adaptation
C) Difference threshold
D) Response bias

10 Jigjiga University · Mid 2012

An audiologist played tones of varying intensities and told the patient to raise her arm whenever she heard a tone. The audiologist was testing:

A) Sensory adaptation
B) Absolute threshold
C) Difference threshold
D) Sensory capability

11 Jimma University · Mid 2015

Given normal vision, a person can detect a candle flame about 48 km away on a dark, clear night. This describes vision's:

A) just noticeable difference
B) absolute threshold
C) sensory adaptation
D) difference threshold

12 ASTU · Mid 2015

"Detection depends on the physical intensity of a stimulus, the sensory process and the observer's decision making." This statement most likely describes:

A) Absolute threshold
B) Difference threshold
C) Depth perception
D) Perceptual constancy

13 Jigjiga University · Mid 2012

When Amelia walks into her mother's kitchen she notices an overpowering smell of onions, but after an hour she no longer complains. This is an example of:

A) Sensory adaptation
B) Just noticeable difference
C) Absolute threshold
D) Psychophysics

14 Model question (Weber's law)

The jnd for weight is about 2 %. A trader holds a 30 kg sack of teff. What is the smallest amount that must be added before he notices the sack is heavier? Would he notice 0.5 kg?

Answers & Explanations

1. B

Sensory adaptation

Decreased responsiveness to an unchanging stimulus is sensory adaptation.

2. A

sensory adaptation

Continuous exposure to the same smell has reduced her sensitivity; the stranger has not adapted.

3. A

Sensory adaptation

Repeated exposure to an unchanging smell leads to sensory adaptation.

4. C

Sensory adaptation

The salt is unchanged; the taste receptors have adapted.

5. False

Adaptation is reduced sensitivity to **unchanging** (constant) stimulation. Changes are exactly what adaptation helps us notice.

6. False

Noticing differences is the role of the difference threshold. Adaptation reduces responses to constant stimuli.

7. False

A high threshold means a stronger stimulus is needed, i.e. lower sensitivity.

8. B

Difference threshold

Noticing a change depends on its size relative to the original (Weber's law), i.e. the difference threshold.

9. C

Difference threshold

The minimum noticeable change is the difference threshold (jnd).

10. B

Absolute threshold

Finding the faintest tone a person can detect measures the absolute threshold.

11. B

absolute threshold

It is the minimum stimulus that can be detected at all.

12. A

Absolute threshold

The module explains that detecting a stimulus depends on intensity and on the person's physical and psychological state (response bias).

13. A

Sensory adaptation

Her sensitivity to the constant smell decreased over time.

14. 0.6 kg; no

$\text{jnd} = 0.02 \times 30 = 0.6$ kg. 0.5 kg is below the jnd, so (more than half the time) he would not notice it.

CHAPTER 2 · LESSON 2.3

Selectivity of Perception: Attention

AFTER THIS LESSON YOU CAN

- Define attention and explain focus and margin
- Explain the external factors that capture attention
- Explain the internal factors, set and motives
- Apply these factors to advertising, teaching and daily life

2.3.1 Attention: focus and margin

D ATTENTION

The perceptual process that **selects certain inputs** for our conscious awareness while **ignoring others**. It is also called **selective perception**.

Our field of experience is divided into a **focus** (what we perceive clearly) and a **margin** (what we perceive dimly). At a football match, the player with the ball is in the **focus**; the other players, the crowd's noise and your cold feet are in the **margin**. When the referee blows the whistle, you suddenly notice how cold your feet are: **attention shifts constantly**, and what was in the margin can move into the focus.

2.3.2 Factors that determine attention

F EXTERNAL (STIMULUS) AND INTERNAL (PERCEIVER) FACTORS

Type	Factor	Meaning	Example
External	**Size and intensity**	bigger, brighter, louder stimuli win	notices in big bold letters; bright clothes
	Repetition	frequently occurring stimuli are noticed	repeated adverts and slogans; a misspelled word repeated many times
	Novelty (contrast)	new or contrasting things stand out	a guest receives special attention; a first-born child
	Movement	moving objects grab attention (they may signal danger)	a moving car vs a parked one
Internal	**Set (expectancy)**	being **ready / primed** for certain input	a husband expecting an important call hears the phone at night; his wife hears the baby cry
	Motives and needs	we notice what satisfies our needs	a hungry student notices the smell of food

★ REPETITION HAS A LIMIT

A stimulus repeated in exactly the same way leads to **adaptation (habituation)**, and we stop noticing it. That is why **novelty** is needed: advertisers repeat a message but keep changing how it looks.

THE SYMBOL "13"

In the row **A 13 C D E** you read the symbol as the letter **B**. In the row **12 13 14 15** you read the same symbol as the number **thirteen**. The stimulus is identical; your **set (expectancy)**, created by the context, changes how you perceive it.

ATTENTION TRAPS

- **Expectancy (set)** is an **internal** factor. Novelty, contrast, repetition, size, intensity and movement are **external**.
- **Moving** objects attract attention **more** than still objects, not less.
- Ignoring classmates' chatter to listen to the lecturer = **selective attention**. Listening to music while doing homework is divided attention, not selective attention.

Exam Practice

10 QUESTIONS · LESSON 2.3

1 Arsi University · Mid 2015

Selective attention is the ability to screen out some sensory information and focus on only a particular portion.

True / False

2 Arsi University · Final 2013

Which of the following individuals is engaging in selective attention?

A) Chala ignores his classmates' talking during his psychology professor's lecture
B) Chaltu listens to music while doing her homework
C) Belete rides his bike while talking on the phone
D) Almaz watches TV while eating

3 Arba Minch University · Mid 2016

Which of the following is NOT an external factor influencing attention?

A) Expectancy
B) Movement
C) Repetition
D) Novelty

4 Addis Ababa Science and Technology University · Mid 2015

All of the following affect selective attention, but one differs from the other three. Which one?

A) Novelty
B) Contrast
C) Expectations
D) Repetition

5 Addis Ababa Science and Technology University · Mid 2015

Which statement about attention is FALSE?

A) People who wear brightly coloured clothes tend to capture your attention
B) New objects tend to capture attention more than familiar objects
C) Non-moving objects tend to get your attention more than moving objects
D) Repeated stimuli are more likely to be noticed

6 Injibara University · Mid 2014

Brighter stimuli are more likely to capture attention than dimmer ones. This refers to which factor?

A) Novelty
B) Repetition
C) Intensity
D) Movement

7 Wolaita Sodo University · Mid 2014

Set refers to the idea that you may be ready and primed for certain kinds of sensory input.

True / False

8 Arba Minch University · Mid 2013

The influence of perceptual set shows that our perception is:

- **A)** at least partly dependent on seeing the world
- **B)** minimally dependent on experience
- **C)** at least partly dependent on our expectations about the world
- **D)** independent of our motives

9 Arsi University · Final 2013

Whether a sensory input becomes figure or ground depends on our attention.

True / False

10 Model question (module activity)

You and your friend are in the same room. You listen to the music while your friend listens to people talking outside. The best explanation is:

- **A)** sensory adaptation
- **B)** differences in set and motives between the two of you
- **C)** different absolute thresholds
- **D)** perceptual constancy

Answers & Explanations

1. True

That is the definition of attention (selective perception).

2. A

Chala ignores his classmates' talking during his psychology professor's lecture

Chala selects one input (the lecture) and screens out the rest. The others are dividing attention between tasks.

3. A

Expectancy

Expectancy (set) is an internal, psychological factor of the perceiver.

4. C

Expectations

Expectation is internal; the others are characteristics of the stimulus (external).

5. C

Non-moving objects tend to get your attention more than moving objects

Moving objects capture attention more than still ones.

6. C

Intensity

Size and intensity (brightness, loudness) are external factors of attention.

7. True

Set (expectancy) is the internal readiness to perceive certain inputs.

8. C

at least partly dependent on our expectations about the world

Set means we perceive partly what we expect to perceive.

9. True

The factors that determine attention also determine what becomes the figure (focus) and what becomes the ground (margin).

10. B

differences in set and motives between the two of you

The stimuli are identical; what each person attends to depends on internal factors (expectancy, interests and needs).

CHAPTER 2 · LESSON 2.4

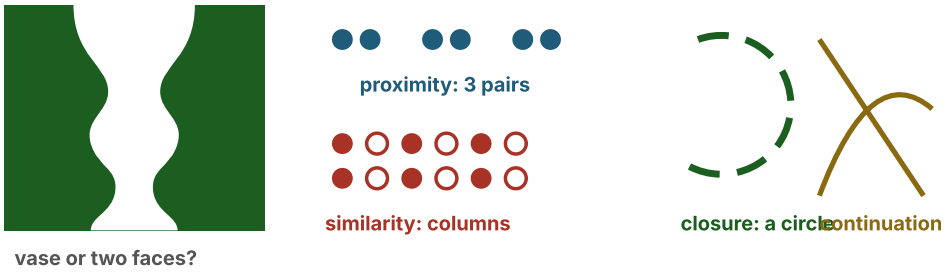
Form Perception and Gestalt Organization

AFTER THIS LESSON YOU CAN

- Explain figure–ground perception and contours
- Apply the Gestalt laws of proximity, similarity, good figure, continuation and closure
- Explain camouflage with contours

2.4.1 Figure and ground

To perceive a **form** (a meaningful shape), we must separate a **figure** (the object) from the **ground** (its background): a picture on a wall, words on a page, a melody over background chords. Figure–ground perception is **active**: the figure and ground can **reverse** (as in the famous vase/two-faces picture).



Figure–ground reversal and the Gestalt laws of organization

2.4.2 Contours

We separate figure from ground because we perceive **contours**: boundaries formed where there is a **marked change in brightness or colour**. A gradual change gives no contour. When contours are **disrupted**, as in **camouflage**, objects are hard to separate from the background.

CONTOURS IN REAL LIFE

- A **chameleon** changes colour to match its surroundings, so no contour separates it from the background and predators (and people) cannot see it.
- Soldiers wear **green** uniforms in forests for the same reason.
- Charcoal writing on a blackboard is nearly invisible; white chalk on a blackboard is clear. (White writing on a white board would also fail.)

2.4.3 Gestalt laws of organization

Gestalt psychologists showed that we organize parts into wholes: "**the whole is greater than the sum of its parts**". Without organization, a face on TV would be just dots and lines.

THE LAWS OF PERCEPTUAL ORGANIZATION

Law	We tend to...	Example
Proximity (nearness)	group items that are **close together** in space or time	stars seen as three groups; a boy and girl sitting together seen as a "couple"
Similarity	group items that **look alike**	in the figure, filled and open dots form columns
Good figure (symmetry)	organize parts into a **balanced, symmetrical** whole	dots and rings seen as one six-pointed star
Continuation	see a line as **continuing** in the same direction	crossing lines seen as one straight and one curved line
Closure	**fill in gaps** to see a **complete** object	a broken circle seen as a circle; understanding a phone call that keeps cutting out

SOLVING A GESTALT QUESTION (MEKELLE 2011)

"Mehari phones his wife in the USA. The line keeps breaking up, yet he understands what she is saying." Which principle?

Step 1: Is the question about **grouping** (nearness, likeness) or about **missing information**? Here parts of the message are missing.

Step 2: Filling in missing parts to perceive a complete whole = **closure**.

Contrast: a teacher who treats students sitting together as a group uses **proximity**; seeing all students in the same uniform as one team is **similarity**.

GESTALT TRAPS

- "Closure = perceiving nearby objects as belonging together" is **false**; that is proximity.
- "Continuation = closing gaps" is **false**; that is closure.
- "The whole is greater than the sum of its parts" does **not** mean the parts are useless; it means new properties emerge from their organization. (The taste of a single spice is a part, not a whole.)
- Form perception applies to the **two-dimensional** world; judging distance is depth perception.

Exam Practice

12 QUESTIONS · LESSON 2.4

1 Arsi University · Mid 2015

The law of perceptual organization that fills in gaps in stimulation is the law of:

A) similarity
B) proximity
C) closure
D) continuation

2 Arsi University · Final 2013

Mr. Gemechu tends to see any boy and girl sitting together as a "couple", whether or not they are dating. This illustrates the Gestalt principle of:

A) Closure
B) Continuity
C) Similarity
D) Proximity

3 Mekelle University · Mid 2011

Mehari often phones his wife in the United States. The line keeps breaking up because of the distance, yet he understands what she says. This illustrates the principle of:

A) Proximity
B) Similarity
C) Closure
D) Continuity

4 Addis Ababa Science and Technology University · Mid 2015

Some animals protect themselves from enemies by matching their colour to their surroundings. Which perceptual organization is being defeated?

A) Closure
B) Figure–ground
C) Similarity
D) Proximity

5 ASTU · Mid 2015

A chameleon changes its colour to match its environment, so people cannot easily see it; similarly, writing with a black marker on a blackboard is hard to read. This happens:

A) when there is a contour in form perception
B) when contours are visually disrupted in form perception
C) when figure–ground is clear
D) when closure is disrupted

6 ASTU · Mid 2015

White writing is more visible on a blackboard than on a white board. Which law of perception does this show?

A) Shape perception
B) Figure–ground perception
C) Grouping
D) Similarity

7 Arba Minch University · Mid 2013

The law of continuation states that we tend to close gaps and perceive meaningful shapes.

True / False

8 Mekelle University · Mid 2016

Closure is the principle by which we perceive objects that are close together in space as belonging together.

True / False

9 Arba Minch University · Mid 2013

The ability of our mind to combine meaningless sensations into meaningful patterns is known as depth perception.

True / False

10 Arba Minch University · Mid 2017

Which of the following is NOT an example of "the whole is greater than the sum of its parts"?

A) The taste of a single spice in a dish
B) The joy experienced at a concert
C) Understanding a complex story
D) Appreciating a beautiful painting

11 Jimma University · Mid 2014

The motto "the whole is greater than the sum of its parts" means the parts are useless.

True / False

12 Jigjiga University · Mid 2014

The capacity to create a complete picture by connecting scattered figures is called:

A) Continuity
B) Similarity
C) Closure
D) Proximity

Answers & Explanations

1. C

closure

Closure fills in missing parts to perceive complete objects.

2. D

Proximity

He groups the two people because they are near each other.

3. C

Closure

His mind fills in the missing parts of the message.

4. B

Figure–ground

Without a contour, the animal (figure) cannot be separated from its background (ground).

5. B

when contours are visually disrupted in form perception

Contours depend on a marked change in brightness or colour; camouflage removes it.

6. B

Figure–ground perception

The sharp contrast creates a contour that separates the figure (writing) from the ground (board).

7. False

Closing gaps is closure. Continuation is perceiving a line as continuing in the same direction.

8. False

That is proximity.

9. False

Organizing sensations into meaningful shapes is form perception. Depth perception judges distance.

10. A

The taste of a single spice in a dish

A single spice is one part. The others are wholes whose meaning emerges from many parts organized together.

11. False

The parts matter, but their organization creates new properties that the parts alone do not have.

12. C

Closure

Completing an incomplete picture is closure.

CHAPTER 2 · LESSON 2.5

Depth Perception, Constancy and Illusion

AFTER THIS LESSON YOU CAN

- Explain binocular cues (retinal disparity, convergence)
- Explain monocular cues including pictorial cues
- Describe size, shape and brightness constancy
- Explain illusions such as the moon and Müller-Lyer illusions, and the claim of ESP

2.5.1 Depth perception

We live in a 3-D world, but the image on the retina is flat. **Depth perception** is the ability to judge **distance**. It uses two kinds of cues.

DEPTH CUES

Type	Cue	How it works
Binocular (need **both** eyes)	**Retinal disparity**	each eye sees a slightly different image; the **closer** the object, the **greater** the disparity
	Convergence	the eyes turn **inward** to focus; more muscle tension = closer object (alcohol disrupts it)
Monocular (one eye is enough)	**Accommodation**	the lens changes shape to focus; more change = closer object
	Motion parallax	when we move, **near** objects seem to pass **faster** than far objects (trees beside the road vs a distant farmhouse)
Pictorial (monocular cues used by artists)	**Interposition** (overlap)	an object that blocks another is closer
	Relative size	of two objects of similar size, the one with the smaller image is farther away
	Linear perspective	parallel lines (railway tracks) seem to **converge** in the distance
	Elevation	objects higher in the visual field seem farther
	Shading	shadows give shape and depth
	Aerial perspective	near objects are **clearer**; distant mountains look hazy
	Texture gradient	near surfaces show **detail**; far surfaces look smooth and dense

PEOPLE WITH ONE EYE STILL PERCEIVE DEPTH

They lose the **binocular** cues (retinal disparity, convergence), but they can still use **monocular** cues: accommodation, motion parallax and all the pictorial cues. "People with one eye detect depth by convergence" is **false**.

SOLVING A DEPTH-CUE QUESTION (AASTU 2015)

"Travelling by bus from Addis Ababa to Debre Berhan, you look out of the window and the nearby trees seem to rush backward faster than the distant hills." Which cue?

Step 1: Is the observer **moving**? Yes → think motion parallax.

Step 2: Near objects appear to move faster than far ones → **motion parallax** (monocular).

Compare: a striped shirt whose lines you cannot make out from far away but can see up close → **texture gradient** (detail). Railway lines meeting in a painting → **linear perspective**. A teacher standing in front of the blackboard → **interposition**.

2.5.2 Perceptual constancy

D PERCEPTUAL CONSTANCY

Perceiving objects as **stable** in size, shape and brightness (and colour) even though the image they produce on the retina changes. It makes the world stable and easier to live in.

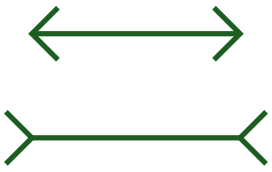
- **Size constancy:** a friend walking away does not seem to shrink, though her retinal image gets smaller. (Alcohol disrupts it.)
- **Shape constancy:** a door or book seen at an angle still looks rectangular.
- **Brightness (and colour) constancy:** a white shirt looks white in dim light or bright sun; your blue bag looks blue under different lights.

2.5.3 Perceptual illusions

D ILLUSION

A **false or incorrect perception** of a real external stimulus. (A **hallucination** is a perception with **no** real stimulus; a **delusion** is a false **belief**.)

- **Rope seen as a snake, a tree seen as a man at night** → illusion (a real stimulus misinterpreted).
- **Moon illusion:** the moon looks larger at the horizon than overhead, though its retinal image is the same size. The horizon cues make it seem farther, so size constancy makes it look bigger.
- **Müller-Lyer illusion:** two equal lines look unequal because of the arrow-like fins at their ends; one looks like the far inside corner of a room, the other like the near outside corner of a building.
- **Edgar Allan Poe's "The Sphinx":** a man sees a "monster" on a distant mountain; it was a small insect on his window. He misapplied size constancy by assuming the insect was far away.



Which line is longer?

Measure them:
they are equal.

The Müller-Lyer illusion

Extrasensory perception (ESP) is the claim of knowing things **without** sensory input: **telepathy** (reading minds), **clairvoyance** (perceiving events happening elsewhere), **precognition** (knowing the future) and **psychokinesis** (moving objects with the mind). Most scientists and psychologists do **not** accept ESP because it has not been supported by controlled research.

! DEPTH, CONSTANCY AND ILLUSION TRAPS

- **Convergence** and **retinal disparity** are **binocular**; **accommodation** is **monocular** (not pictorial).
- **Overlap (interposition)** and linear perspective are pictorial cues; convergence and accommodation are not.
- Retinal disparity is **greater** for **closer** objects.
- In aerial perspective, **nearer** objects look **clearer**.
- Illusion = real stimulus misread; hallucination = no stimulus at all.

Exam Practice

18 QUESTIONS · LESSON 2.5

1 Arsi University · Mid 2015

People with one eye can perceive depth by means of convergence.

True / False

2 Addis Ababa Science and Technology University · Mid 2015

Travelling by bus from Addis Ababa to Debre Berhan, you watch outside and notice that nearby trees seem to pass backward quickly, although trees do not move. This is:

A) Accommodation
B) Figure–ground
C) Motion parallax
D) Interposition

3 Bahir Dar University · Final 2015

A person is wearing a white shirt with lines and dots. From far away you cannot make out the lines and dots, but when he comes closer you can. This illustrates:

A) Interposition
B) Linear perspective
C) Texture gradient
D) Relative size

4 Ambo University · Mid 2014

Judging an object's distance from how much our eyes turn inward to focus on it is called:

A) retinal disparity
B) convergence
C) accommodation
D) elevation

5 Bahir Dar University · Final 2015

Which one of the following is a non-pictorial cue?

A) Relative size
B) Elevation
C) Retinal disparity
D) Linear perspective

6 Mekelle University · Mid 2016

The **__ an object is to the observer, the** ____ the retinal disparity.

A) closer; greater
B) closer; smaller
C) farther; greater
D) farther; equal

7 Jimma University · Mid 2015

When we look at an object, each eye receives a slightly different image. This cue is called **__, and it is a** ____ depth cue.

A) convergence; binocular
B) accommodation; monocular
C) retinal disparity; binocular
D) linear perspective; monocular

8 Jimma University · Mid 2015

In a painting, railway tracks seem to go off into the distance because the artist draws them as converging lines. Which depth cue is used?

A) Interposition
B) Texture gradient
C) Relative size
D) Linear perspective

9 Mekelle University · Mid 2016

In which monocular cue does one object blocking the view of another indicate distance?

A) Interposition
B) Texture gradient
C) Linear perspective
D) Aerial perspective

10 Arba Minch University · Mid 2016

In aerial perspective, nearer objects are perceived as clearer.

True / False

11 Addis Ababa University · Mid 2014

As a friend walks away down the street, the image on your retina gets smaller and smaller, yet you do not perceive her as shrinking. This illustrates:

A) shape constancy
B) size constancy
C) location constancy
D) brightness constancy

12 Ambo University · Mid 2014

Mr. X perceives his blue bag as the same colour under different lighting conditions. This is:

A) Colour constancy
B) Size constancy
C) Shape constancy
D) Depth constancy

13 Bahir Dar University · Final 2015

The image of an object on your retina may vary in size, shape and brightness, yet you continue to perceive it as stable. This characteristic of perception is:

A) Perceptual constancy
B) Form perception
C) Selectivity of perception
D) Depth perception

14 Arba Minch University · Mid 2016

Perceiving a rope as a snake at night is an example of:

A) Perceptual organization
B) Perceptual illusion
C) Perceptual constancy
D) Perceptual attention

15 Wolaita Sodo University · Mid 2014

A false interpretation of physical reality is referred to as:

A) Illusion
B) Perception
C) Hallucination
D) Delusion

16 ASTU · Mid 2015

The moon looks larger at the horizon than overhead, although its retinal image is the same size. This is a __ **produced by the misapplication of _, *with the horizon cues acting through* ___.**

A) visual illusion; size constancy; relative size
B) relative size; visual illusion; size constancy
C) visual illusion; relative size; size constancy
D) size constancy; relative size; visual illusion

17 Ambo University · Mid 2014

A friend claims she can know what is happening at her home while she is at university, without any sensory contact. Such a claim is an example of:

A) Perceptual selectivity
B) Perceptual illusion
C) Extrasensory perception
D) Perceptual constancy

18 Arba Minch University · Mid 2016

The perception of an event happening somewhere else without sensory input is called:

A) Telepathy
B) Precognition
C) Psychokinesis
D) Clairvoyance

Answers & Explanations

1. False

Convergence needs two eyes. One-eyed people use monocular cues such as accommodation, motion parallax and pictorial cues.

2. C

Motion parallax

When the observer moves, nearby objects seem to pass faster than distant ones.

3. C

Texture gradient

Near surfaces show more detail; far surfaces look smooth.

4. B

convergence

Convergence is the inward turning of both eyes; more convergence means a closer object.

5. C

Retinal disparity

Retinal disparity is a binocular cue. The others are pictorial cues artists use.

6. A

closer; greater

Near objects give the two eyes very different views.

7. C

retinal disparity; binocular

Retinal disparity comes from the two eyes' different views, so it is binocular.

8. D

Linear perspective

Parallel lines appearing to meet in the distance is linear perspective.

9. A

Interposition

The blocking (overlapping) object is perceived as closer.

10. True

Distant objects look hazy because of the air between them and us.

11. B

size constancy

Size constancy interprets a smaller retinal image as greater distance, not a smaller person.

12. A

Colour constancy

Perceiving a stable colour despite changes in lighting is colour (brightness) constancy.

13. A

Perceptual constancy

That is the definition of perceptual constancy.

14. B

Perceptual illusion

A real stimulus (the rope) is misinterpreted.

15. A

Illusion

An illusion misreads a real stimulus. A hallucination has no stimulus; a delusion is a false belief.

16. A

visual illusion; size constancy; relative size

The moon illusion is a visual illusion caused by misapplied size constancy; the horizon's objects (relative size cues) make the moon seem farther and therefore larger.

17. C

Extrasensory perception

Knowing without the senses is the claim of ESP (here, clairvoyance).

18. D

Clairvoyance

Clairvoyance = perceiving distant events. Telepathy = reading minds; precognition = knowing the future; psychokinesis = moving objects with the mind.

CHAPTER 2 · QUICK REVISION

Sensation and Perception — at a glance

CHAPTER 2 AT A GLANCE

Topic	Must-know points
Sensation	detection/stimulation by receptors; stimulus = energy; transduction = energy → neural impulse
Perception	organization and interpretation; needs sensation; prosopagnosia case
Thresholds	absolute (minimum detected 50 % of the time); difference / jnd (minimum change); Weber: jnd = constant fraction; lower threshold = more sensitive
Adaptation	less response to an unchanging stimulus; not to extreme pain or cold
Attention	focus and margin; shifts constantly; external: size/intensity, repetition, novelty, movement; internal: set, motives
Form	figure–ground; contours (camouflage); Gestalt: proximity, similarity, good figure, continuation, closure
Depth	binocular: retinal disparity, convergence; monocular: accommodation, motion parallax, interposition, relative size, linear perspective, elevation, shading, aerial perspective, texture gradient
Constancy	size, shape, brightness/colour
Illusion	moon, Müller-Lyer, rope-snake; ESP not scientifically supported

GESTALT LAWS: "PLEASE SHOW GOOD COMMON SENSE"

Proximity, **S**imilarity, **G**ood figure, **C**ontinuation, **C**losure.